Oil ring calculation

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1. Purpose

This document defines the guidelines to for the oil ring dimensioning.

2. Scope.

This procedure is applicable to oil ring dimensioning, and it is based on empirical results in literature.

The procedure is applicable for the Nova Milanese and La Spezia plants.

3. Abbreviations.

c_r	journal radial clearance [in]
d	journal diameter [in]
D	inside diameter of oil ring [in]
I	depth of oil ring inner diameter beneath the static oil level [in]
L	bearing width [in]
n	journal rotational speed [rps]
N	oil ring rotational speed [rps]
n_f	journal rotational speed (full film) [rps]
N_f	oil ring rotational speed (full film) [rps]
n_s	journal rotational speed (start of slip) [rps]
N_s	oil ring rotational speed (start of slip) [rps]
n_t	journal rotational speed (transition) [rps]
N_t	oil ring rotational speed (transition) [rps]
p	oil ring width [in]
Q	oil ring delivery [in^3/s]
Q_{min}	minimum required oil ring delivery [in^3/s]
Q_t	oil ring delivery at transition speed V_t [in^3/s]
v	journal surface velocity [ips]
V	oil ring internal diameter surface velocity [ips]
W	weight of oil ring [lb]
μ	oil viscosity [Reyns]
ν	oil kinematic viscosity [in^2/s]

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4. Basic knowledge: bearing oil ring performances

Oil ring working principia is relatively simple. The rotation of the ring depends on a propulsive force between the rotating shaft and the ring. At low speeds the static friction between the ring and journal is sufficiently high for the ring to be driven with its surface speed equal to that of the journal. At higher speeds slip occurs and the driving force is transmitted through an oil film which is analogous in many ways to that in a journal bearing.

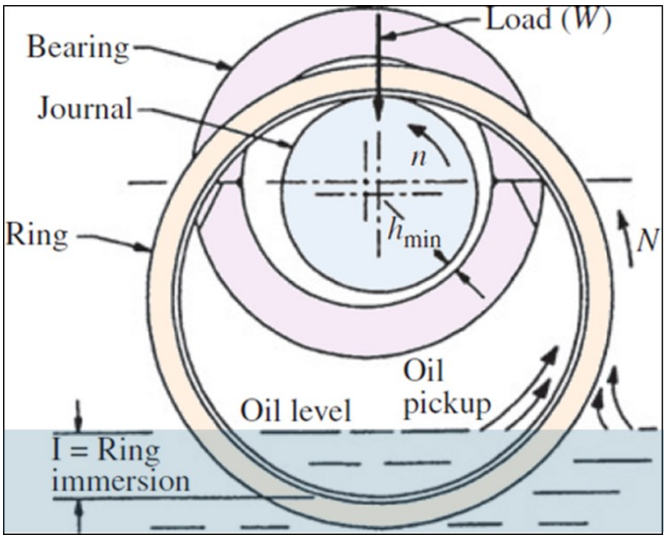

Figure 1 Oil ring.

Modifications of the geometry of the inside circumference of the ring have been used in some cases to develop a higher friction coefficient and thus a more positive drive. Opposing the ring rotation is the drag of the lower portion of the ring in the oil reservoir, the force required to lift the oil up from the reservoir to the top of the journal, and the frictional drag on the ring by any close-running stationary surfaces such as the sides of the ring slot in the bearing.

Oil delivery mechanisms is more complicated. The oil lifted from the oil reservoir as a coating has been observed to be delivered to the journal largely from the inside surface of the ring by a squeezing action as the ring passes over the rotating journal. At low speeds some oil from the sides of the ring was found to be available, but this quantity dropped off rapidly as the centrifugal force threw out the oil at higher speeds. In the highest speed range of operation, the centrifugal force throws off as a spray most of the oil lifted from the reservoir.

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5. Factors affecting oil delivery

5.1. Depth of immersion

Usually, an increase in submersion causes the decrease of ring speed and oil delivered. Anywhere if the ring is sufficiently below the level of the oil in the reservoir that the inner surface is never exposed at any speed, oil delivery is not drastically changed with reasonable changes in oil level. For the variation in level experienced in bearings in actual service, i.e., from a minimum immersion of about 10%t of ring diameter to the maximum immersion of about 20%, the change in oil delivery found in these tests was always less than 20% and generally about 10%.

5.1. Oil viscosity

The effect of changes in oil viscosity on ring speed and oil delivery has also been extensively studied: as the temperature of the oil increases, its viscosity decreases, and the quantity of lubricant delivered by the ring decreases accordingly. The viscosity of the oil affects the speed of the ring to a much lesser extent than it affects the oil delivery.

5.2. Oil ring speed

The most important factor affecting the oil deliver of an oil ring is the speed of the ring itself. As the speed of the ring is increased, the quantity of oil delivered increases up to a point. When the velocity of the ring becomes too large, centrifugal force flings the oil off the ring and it is lost for lubrication purposes. Table 1 summarises the empirical formulas useful for these calculations.

Table 1 Relationship between oil ring deliver and oil ring speed.

	Oil delivering equations	
Low ring speed	$Q = 0.14pv^{0.65}(DN)^{1.5}$	
Transition speed	$N_t = 0{,}063\frac{D^{0,5}}{v^{0.67}}$	$V_t = \pi N_t D$
High ring speed	$Q = Q_t\frac{N}{N_t}\left(\frac{n_t}{n}\right)^{0,5}$	$Q_t = 0.14pv^{0.65}(DN_t)^{1.5}$

A transition ring speed V_t exists above which oil delivery no longer increases in the manner described in the first equation A likely cause of this transition in delivery is the increased centrifugal

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action which reduces availability to the journal of oil exuded from between the ring and journal. Above this transition speed, oil delivery increases much less rapidly with increasing ring speed.

6. Relationship between ring speed and journal speed

The variation in ring speed observed with variations in journal speed almost always shows three distinct regions:

- at low journal speeds the velocity of the inside circumference of the ring is identical to that of the outside circumference of the journal (**no slip region**);
- during an increase of journal speed through this no-slip region, the static friction between the ring and journal is no longer sufficient to maintain synchronous ring speed; beyond this journal speed a region of slower ring rotation is encountered in which the drive of the ring is partially through metallic contact and partially by a fluid film (**partial film drive**);
- as the journal speed goes still higher, a full-film region is finally reached in which the ring is driven entirely by hydrodynamic action through an oil film (**full film drive**).

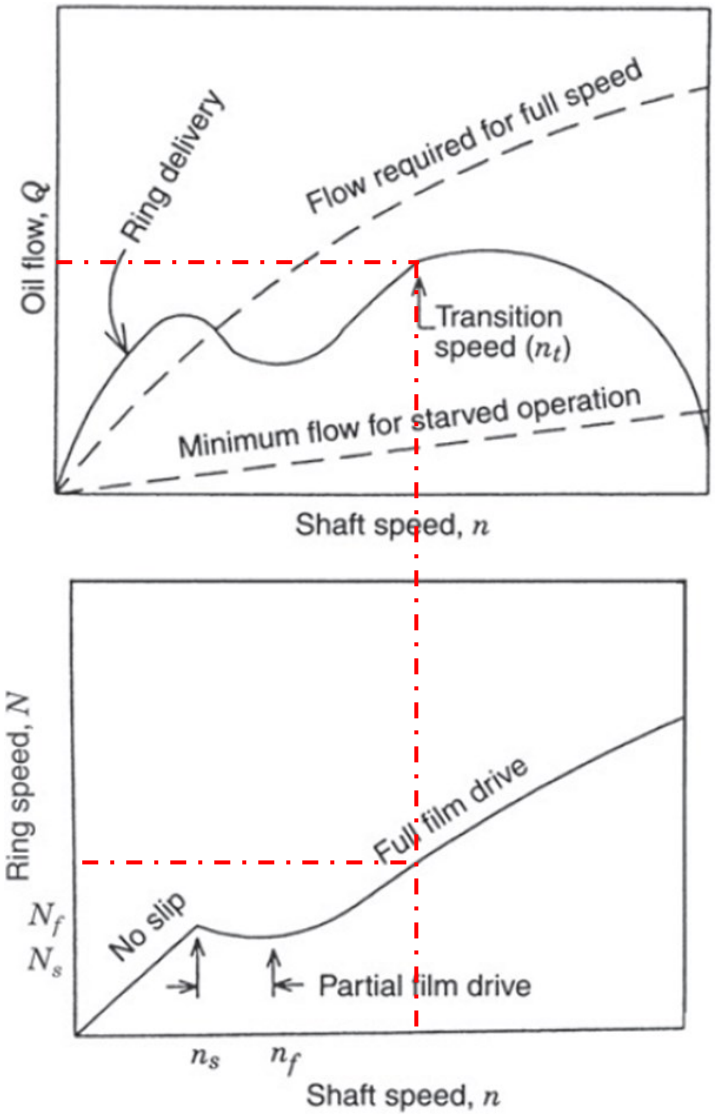

Figure 2 Oil ring speed and oil flow.

Figure 2 illustrates the three different regions for the ring speed; moreover, the oil flow and the transition speed are shown in the upper part of the figure.

Equations governing the ring speed are summarised in Table 2 and Table 3.

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Table 2 Relationship between journal and oil ring speed.

	Ring speed equations			
	oil ring		journal	
No-slip drive	$N = n\frac{d}{D}$	$V = v$	-	-
Start of slip	$N_s = 0{,}0048\frac{W}{\mu D^2}$	$V_s = \pi N_s D$	$n_s = N_s\frac{D}{d}$	$v_s = V_s$
Start of full film operation	$N_f = k_f\frac{\nu^{0.2}}{D^2}(nd^2)^{0.8}$	$V_f = \pi N_f D$	$n_f = 2.9n_s$	$v_f = 2.9v_s$

The value of k_f in N_f formula is function of the ring immersion I[1] in accordance with Table 3.

Table 3 k_f values.

$\frac{I}{D}$ [−]	5%	10%	15%	20%	25%
k_f [−]	2.08	1.81	1.67	1.57	1.51

[1] A ratio I/D from 10% to 15% is suggested.

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7. Partially starved oil-film bearings

For an oil ring journal bearing, a requisite for satisfactory operation is an adequate lubricant supply. Required oil flow for the bearing can be calculated with specialized tools (e.g: BePerf, Madyn Alp3t, etc…) or estimated by formula:

$$Q_{min} = 0{,}039 \times ndL(2c_r)$$

For several applications bearings involves a smaller lubricant supply than is needed to be fully flooded. Three general realms are suggested in Table 4 for the degree of lubricant feed for fluid film operation:

- full feed, with all the lubricant being fed that can be accommodated for full hydrodynamic behavior;
- partially starved, to provide an incomplete hydrodynamic fluid film;
- minimal feed to create a fluid film similar in thickness to the surface roughness.

Table 4 Fluid film bearing operation.

	Supply methods	**Typical applications**	**Approximate minimum oil feed rate, % of full-film flow rate**
Full feed	Pressure fed, flooded	Turbines, compressors, superchargers	100
Partially starved	Ring-oil, disc	Electric motors, pumps, fans, line shafts, miscellaneous large machines	10
Minimal feed	Wick, drop feed, mist, air-oil	Appliances, conveyors, pumps, various small and low-speed units	1

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An oil ring should be considered acceptable if it is in accordance with the following characteristics:

- full film operation (preferred[2]);
- journal transition speed greater than journal full film operation;
- oil delivered by the ring greater than four times the minimum required evaluated by formula;
- starvation ratio, i.e. the ratio between oil delivered by the ring and calculated with special toll (e.g. Beperf, Alp3T, etc…), is greater than 40%.

[2] If oil ring speed falls in the transition region, a redesign of the ring must be performed.

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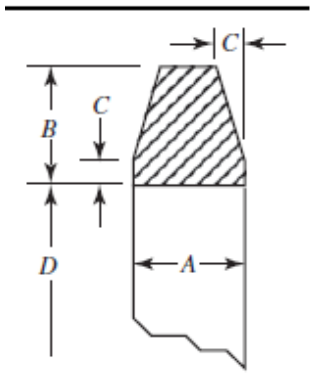
8. Preliminary sizing

Common oil-ring dimensions are indicated in Table 5. Ring slot must be between 120% and 130% of ring width. One of the important elements of the ring profile is its trapezoid configuration. Because of high side drag, either side of the ring can stick to a wall of the bearing slot, causing immediate reduction in ring speed. To reduce the side drag, the side length of the ring (perpendicular to the shaft) should be small (1 or 2 mm for larger rings) and the rest is usually cut. Suggested immersion must be 10% - 15% of ring internal diameter.

Table 5 Typical trapezoidal ring dimension.

d [mm]	D [mm]	A [mm]	B [mm]	C [mm]	
53 -67	100	6	6	1	
71 - 90	135	7	7	1	
87 - 110	165	8	8	2	
105 - 133	200	16	13	2	

9. References

Bearing Oil-Ring Performance - D. Lemon E. Booser

Experimental Study of Stable High-Speed Oil Rings - H. Hesmat O. Pinkus

Performance of Starved Journal Bearings With Oil Ring lubrication - H. Hesmat O. Pinkus

Factors Affecting Oil Ring And Slinger Lubricant Delivery & Stability - S. Bradshaw J. Hawa J. Salerno

Applied Tribology Bearing Design and Lubrication – E. Booser

Mod.500 – Oil ring calculator

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10. Appendix A: Oil ring calculator

A calculation sheet called *"Mod.500 Oil ring calculator"* is provided to evaluate oil ring design. There are five sections illustrated in Figure 3

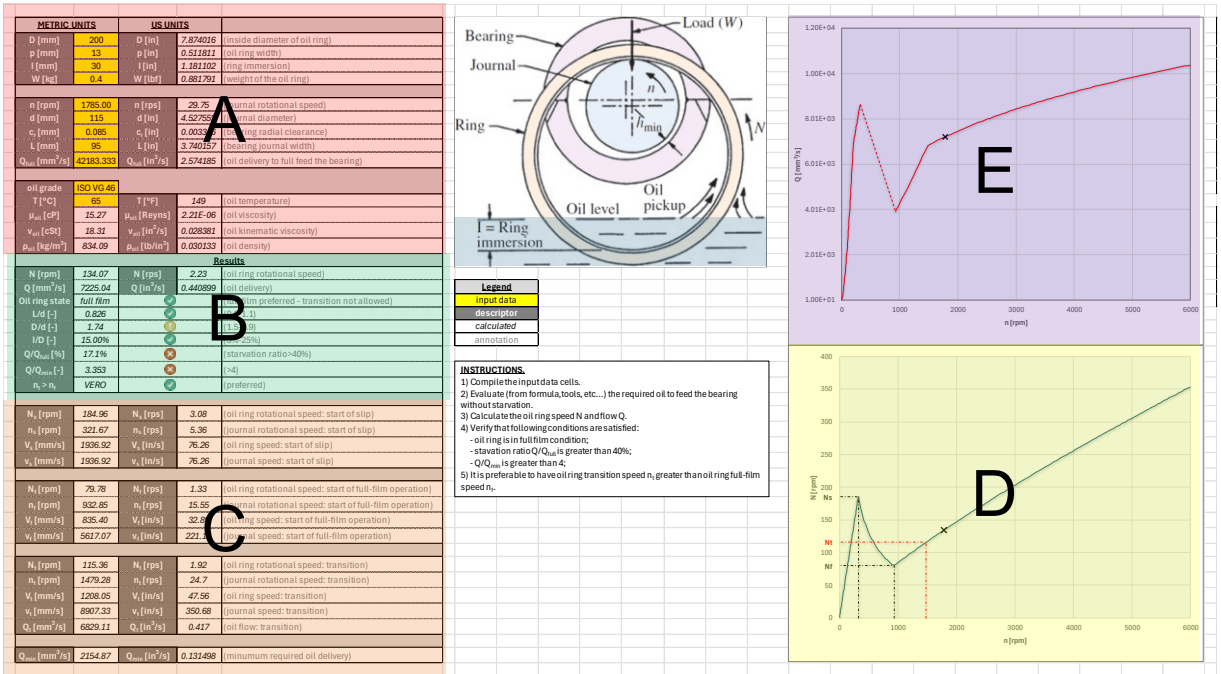

Figure 3 Oil ring calculator.

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10.1. Section A: input data

In this section all the input data required to evaluate the oil ring design are showed (Figure 4):

- inside diameter of oil ring;
- oil ring width;
- ring immersion;
- weight of the oil ring;
- journal rotational speed;
- journal diameter;
- bearing radial clearance;
- bearing journal width;
- oil delivery to full feed the bearing (evaluated from bearing calculations);
- oil type/temperature (required to evaluate oil viscosity and kinematic viscosity).

METRIC UNITS		US UNITS		
D [mm]	200	D [in]	7.874016	(inside diameter of oil ring)
p [mm]	13	p [in]	0.511811	(oil ring width)
l [mm]	30	l [in]	1.181102	(ring immersion)
W [kg]	0.4	W [lbf]	0.881791	(weight of the oil ring)
n [rpm]	1785.00	n [rps]	29.75	(journal rotational speed)
d [mm]	115	d [in]	4.527559	(journal diameter)
c_r [mm]	0.085	c_r [in]	0.003346	(bearing radial clearance)
L [mm]	95	L [in]	3.740157	(bearing journal width)
Q_{full} [mm^3/s]	42183.333	Q_{full} [in^3/s]	2.574185	(oil delivery to full feed the bearing)
oil grade	ISO VG 46			
T [°C]	65	T [°F]	149	(oil temperature)
μ_{oil} [cP]	15.27	μ_{oil} [Reyns]	2.21E-06	(oil viscosity)
ν_{oil} [cSt]	18.31	ν_{oil} [in^2/s]	0.028381	(oil kinematic viscosity)
ρ_{oil} [kg/m^3]	834.09	ρ_{oil} [lb/in^3]	0.030133	(oil density)

Figure 4 Input data.

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10.2. Section B: results

In this section calculated results and control check are summarised (Figure 5):

- oil ring rotational speed;
- oil delivery;
- oil ring state (no slip/transition/full film);
- geometry check on ring dimensions (L/d. D/d and l/D);
- verification of starvation ratio (>40%);
- verification of minimum oil flow required (>4);
- verification of n_t and n_f (it is preferable that journal transition speed would be greater than journal full film speed).

Results				
N [rpm]	134.07	N [rps]	2.23	(oil ring rotational speed)
Q [mm^3/s]	7225.04	Q [in^3/s]	0.440899	(oil delivery)
Oil ring state	full film	✓		(full film preferred - transition not allowed)
L/d [-]	0.826	✓		(0.5-1.1)
D/d [-]	1.74	!		(1.5-1.9)
l/D [-]	15.00%	✓		(5%-25%)
Q/Q_{full} [%]	17.1%	✗		(starvation ratio>40%)
Q/Q_{min} [-]	3.353	✗		(>4)
$n_t > n_f$	VERO	✓		(preferred condition)

Figure 5 Results.

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10.3. Section C: calculated data

In this section calculated data are shown (Figure 6):

- start of slip condition (rotational and peripheral speed of ring and journal);
- full film operation (rotational and peripheral speed of ring and journal);
- transition (rotational and peripheral speed of ring and journal);
- oil ring state (no slip/transition[3]/full film);
- minimum required oil delivery;
- verification of starvation ratio (>40%);
- verification of minimum oil flow required (>4);
- verification of n_t and n_f (it is preferable that journal transition speed would be greater than journal full film speed).

N_s [rpm]	184.96	N_s [rps]	3.08	(oil ring rotational speed: start of slip)
n_s [rpm]	321.67	n_s [rps]	5.36	(journal rotational speed: start of slip)
V_s [mm/s]	1936.92	V_s [in/s]	76.26	(oil ring speed: start of slip)
v_s [mm/s]	1936.92	v_s [in/s]	76.26	(journal speed: start of slip)
N_f [rpm]	79.78	N_f [rps]	1.33	(oil ring rotational speed: start of full-film operation)
n_f [rpm]	932.85	n_f [rps]	15.55	(journal rotational speed: start of full-film operation)
V_f [mm/s]	835.40	V_f [in/s]	32.89	(oil ring speed: start of full-film operation)
v_f [mm/s]	5617.07	v_f [in/s]	221.14	(journal speed: start of full-film operation)
N_t [rpm]	115.36	N_t [rps]	1.92	(oil ring rotational speed: transition)
n_t [rpm]	1479.28	n_t [rps]	24.7	(journal rotational speed: transition)
V_t [mm/s]	1208.05	V_t [in/s]	47.56	(oil ring speed: transition)
v_t [mm/s]	8907.33	v_t [in/s]	350.68	(journal speed: transition)
Q_t [mm^3/s]	6829.11	Q_t [in^3/s]	0.417	(oil flow: transition)
Q_{min} [mm^3/s]	2154.87	Q_{min} [in^3/s]	0.131498	(minimum required oil delivery)

Figure 6 Calculated data.

[3] Transition is an unfavourable condition and should be avoided.

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10.4. Section D: oil ring rotating speed

In this section oil ring rotating speed is described in Figure 7:

- Ns, N_f and N_t are in accordance with formulas in Table 2;
- operating point is highlighted with a X indicator.

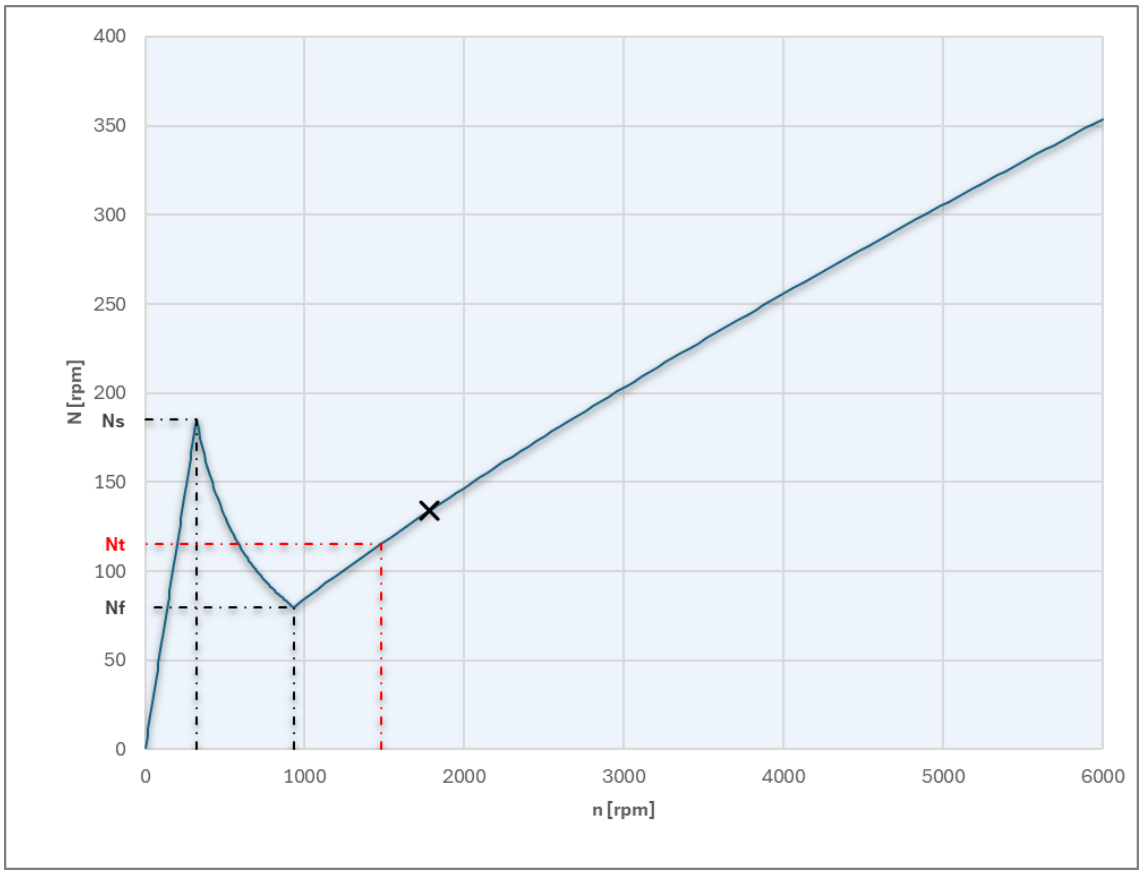

Figure 7 Oil ring rotating speed.

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10.5. Section E: oil delivered

In this section oil delivered by the oil ring is illustrated in Figure 8:

- equations are in accordance with Table 1;
- dotted line shows transition between no slip and full film and must be avoided;
- operating point is highlighted with a X indicator.

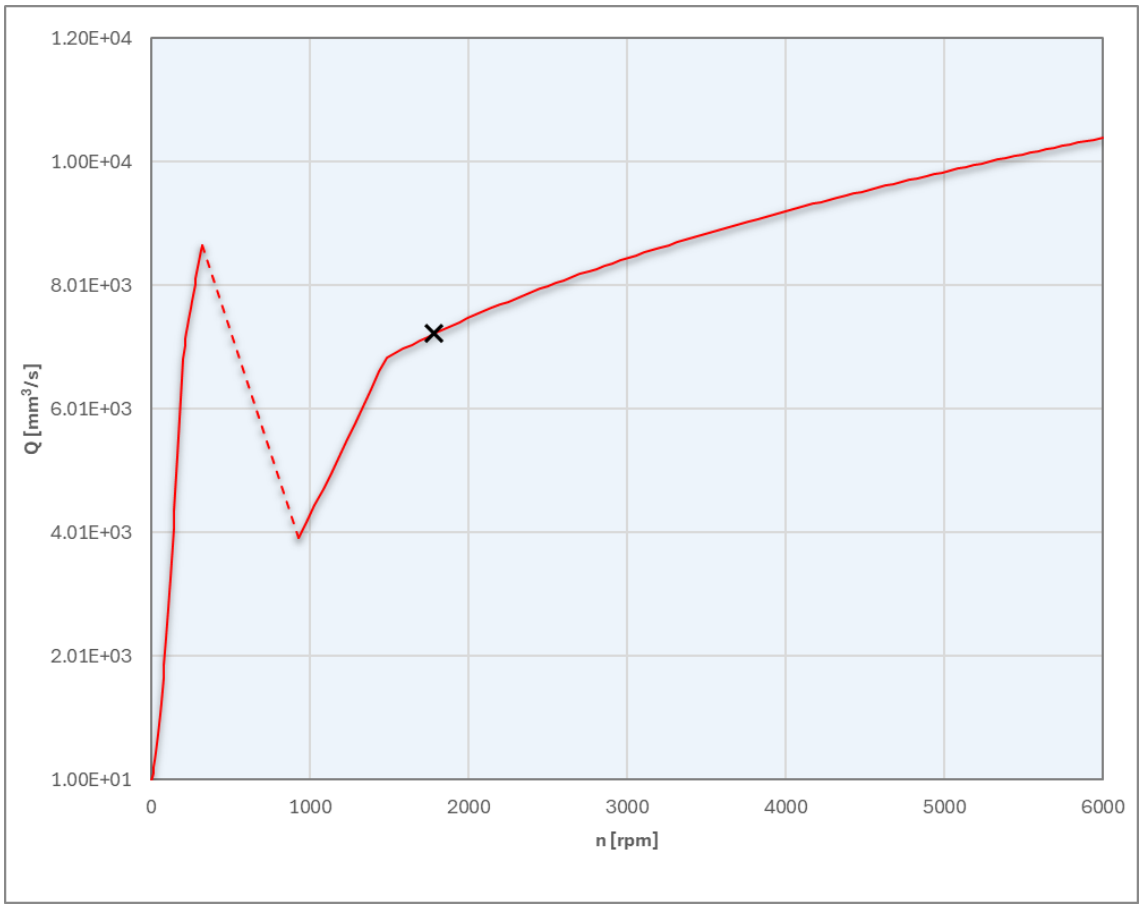

Figure 8 Oil delivered.

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11. Appendix B: Lesson learned

11.1. General discussion

The 300DD2D pump, ref. 23C039 – 086909, is a two-stage pump with sleeve-ball configuration. Both radial/sleeve and thrust/ball bearing have ring-oil feeding system.

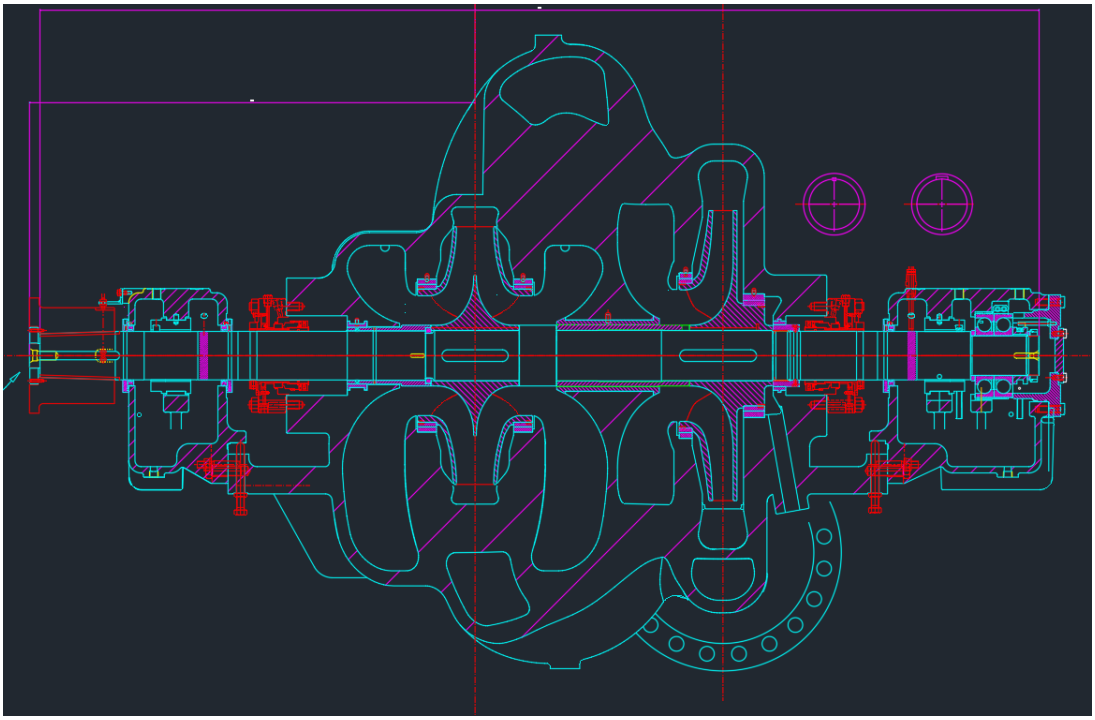

Figure 9 300DD2D pump.

Radial sleeve bearing on DE and NDE side are designed by Osbourne, they are dimensioned on the basis of radial forces calculated by formula in TM.P. The main dimensions of bearing and oil-ring are summarized in Table 7 and Table 6.

Table 6 Oil-ring dimensions.

D_{int} [mm]	*ring internal. diameter*	-	200	**oil ring**
D_{ext} [mm]	ring external. diameter	-	216	
p [mm]	*ring width*	-	13	
l [mm]	*ring immersion*	min	2,5	
		max	12,5	
		average	7,5	

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Table 7 Bearing dimensions.

D_{jou} [mm]	*shaft diameter*	min	115,003	**bearing**
		max	115,025	
		average	115,014	
$D_{bearing}$ [mm]	*bearing diameter*	min	115,17	
		max	115,192	
		average	115,181	
c_{rad} [mm]	*radial clearance*	average	0,0835	
L [mm]	*bearing widht*	up	70	
		low	95	
		average	7,5	

Radial and thrust oil rings have the same dimensions, except that radial oil rings are trapezoidal with one internal groove and axial is rectangular/smooth.

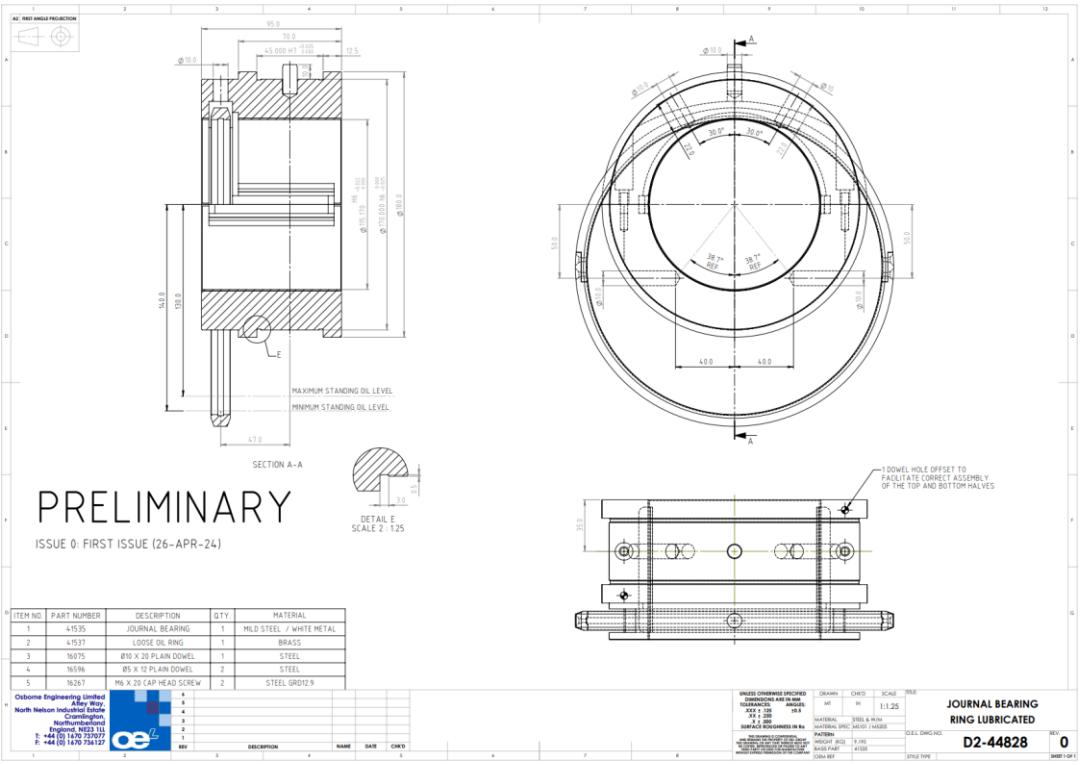

Figure 10 Radial sleeve bearing.

During internal tests, serious damages were found on both radial bearings, probably generated by a *"loss of film"*, whose most common causes are:

- insufficient lubricant supply;
- inappropriate lubricant type (viscosity, temperature, etc…);

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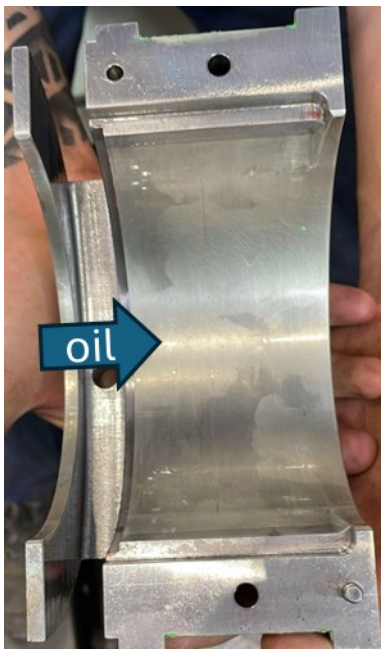
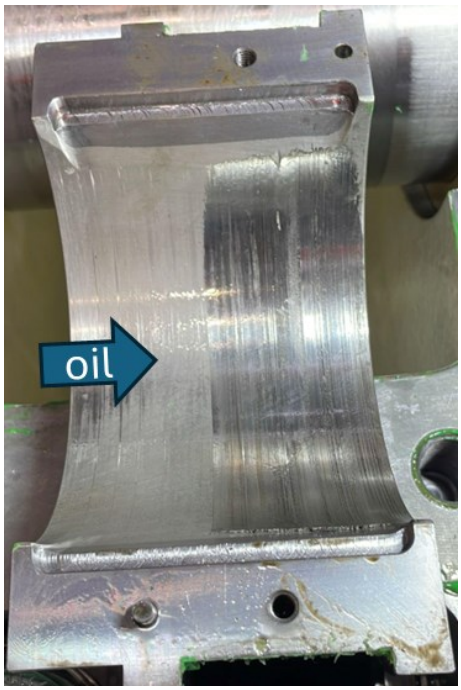
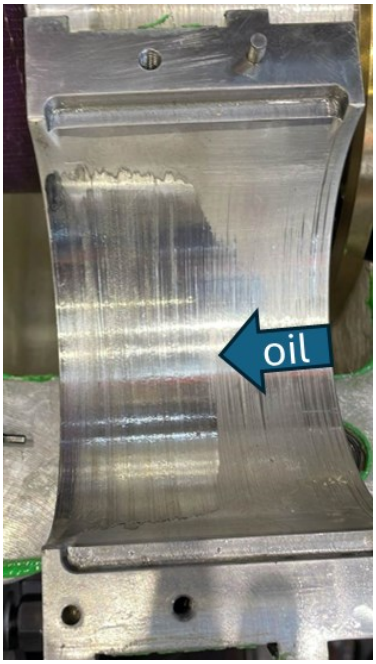
- deflect or distortion of journal pads or collar;
- excessive loads during start-up, transient, etc…

Table 8 Bearing damage.

	DE side	NDE side
Upper (unloaded part)		
Lower (loaded) part		

The oil starvation in the two bearing is evident: the damage is located in the part farther from the oil-ring; it is probable that lubricant supply is not capable to feed in whole length of the baring.

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To solve the malfunction, it was decided to improve the oil feeding adding a second oil-ring to the original design.

After this improvement, significant changes in the rotor/bearing behaviour were found.

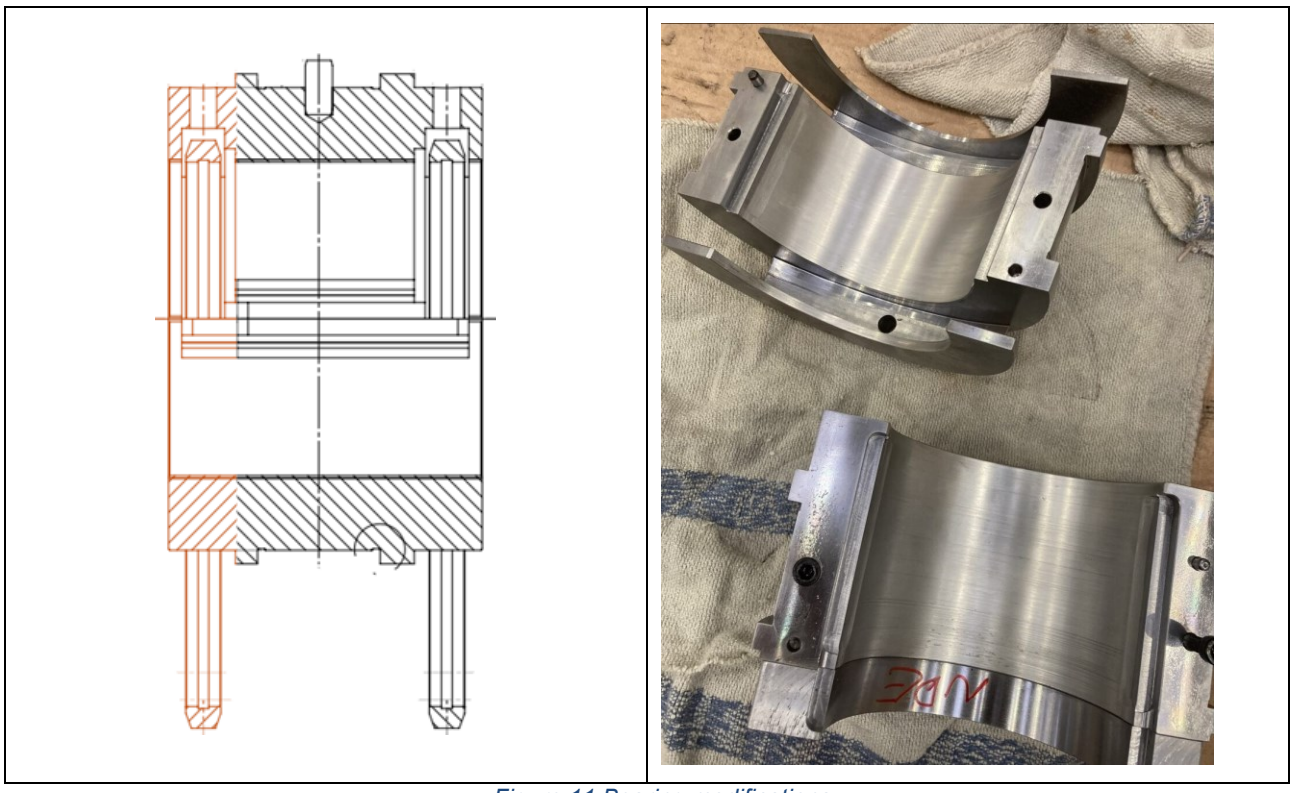

Figure 11 Bearing modifications.

NDE bearing showed superficial scratches, not completely classifiable as *scoring*[4]; moreover, DE bearing has minimum sign of wear. After the test, especially on NDE, oil inspection highlighted some bronze particles, probably caused by the wear of thrust bearing oil-ring. Confirming this hypothesis, the bearing temperatures showed an increasing trend and reached unacceptable levels.

To solve the temperature increase, that could compromise the correct behavior of the lubricant, a fan on NDE side was provided.

To eliminate this undesired oil contamination, that is the most probable cause of the superficial groves on NDE radial bearing, a two-step improvement of the thrust oil-ring carrier was defined and tested (Figure 13):

A. the axial clearance between ring and carrier was increased from 2 mm to 6 mm;

[4] Scoring can be identified as circular parallel grooves on the bearing-lining surface, mainly over the *load zone*, and it is primarily caused by contamination.

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B. the carrier was shaped to reduce the amount of the surface that can became in contact during operation.

Figure 12 Shows the effects of these improvements: with higher clearances the wear on the oil ring and the superficial scratches on the bearing were highly reduced; with the final version the oil-ring has negligible sign of wear, oil contamination was minimum and NDE radial bearing has no sign of scoring.

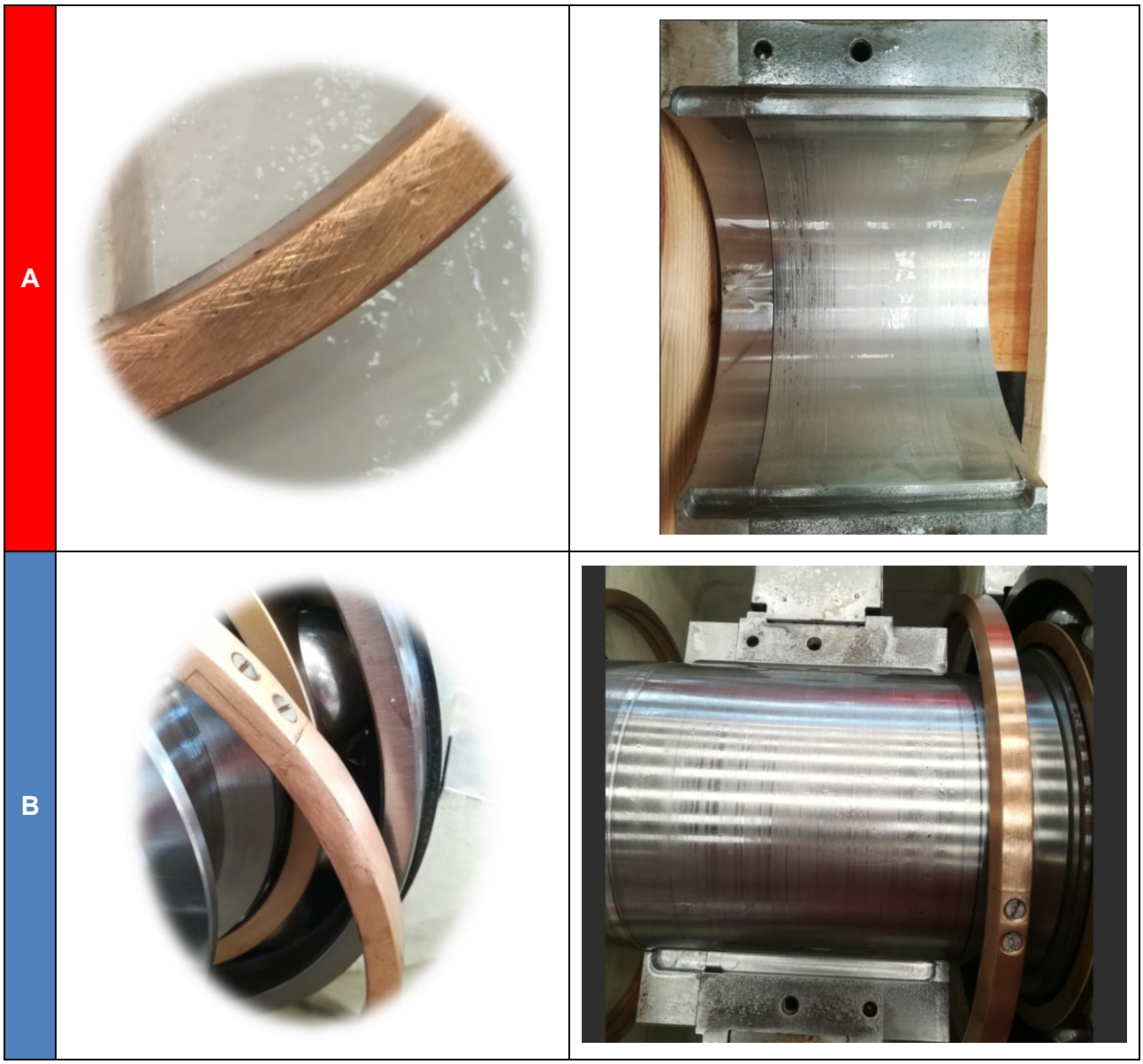

Figure 12 Effect of carrier modification.

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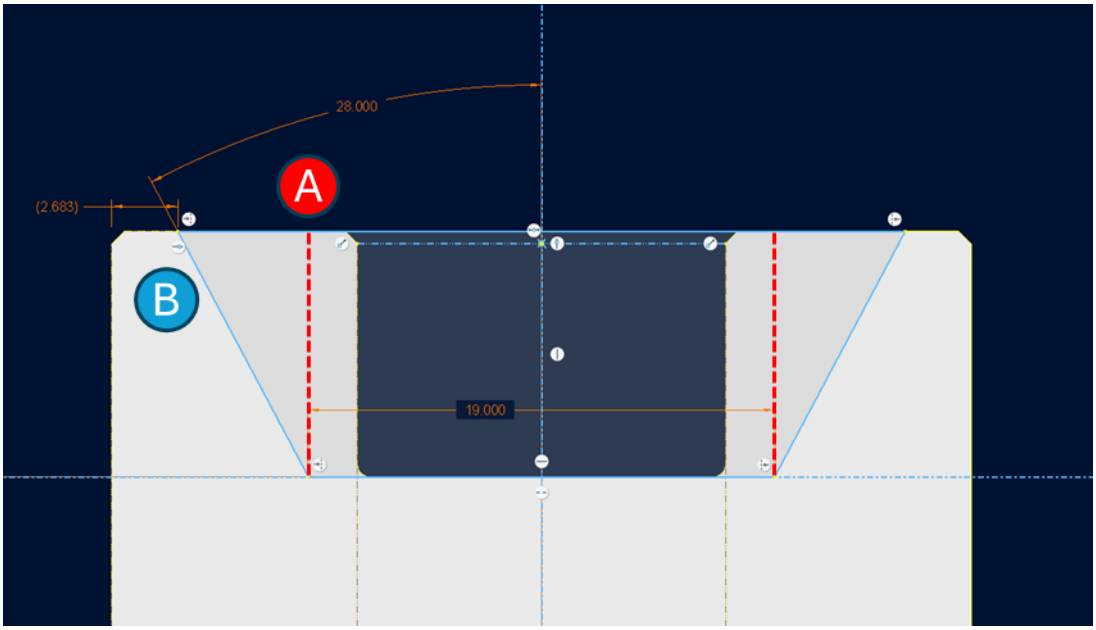

Figure 13 Carrier improvements.

11.2. Conclusions

The adoption of cooling fan combined with the implementation of the previous explained modifications reduced bearing temperature to allowable and stable values and reduce to negligible values the bearing deterioration.

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